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Paleomagnetism of Bhander Sediments from Bhopal Inlier, Vindhyan Supergroup

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Abstract: Paleomagnetic investigations have been carried out on poorly determined radiometric age controls of Bhander sandstones within the vicinity of Bhopal Inlier of the Upper Vindhyan Supergroup. Available ages assigned to the Upper Vindhyan sequence range from Cambrian to the Mesoproterozoic and are derived from a variety of sources and methods. Paleomagnetic data generated from the Bhander Group of Bhopal Inlier yielded a mean declination of 357° and mean inclination of 58° (k=17.69, α_{95} = 16.38) with a Virtual Geomagnetic Pole (VGP) at 74° N, 69.0° E. This pole position is falling close to the Malani Igneous Suite (MIS) mean palaeomagnetic pole of 67.8° N and 72.5° E (A_{95}=8.8°) by Gregory et al. (2009). The results obtained from this study and previous work on the 1073 Ma Majhgawan kimberlite, as well as detrital zircon geochronology of the Upper Bhander sandstone suggest that the Upper Vindhyan sequence may be older than is commonly thought earlier.

Keywords: Paleomagnetism, Malani Igneous Suite, Bhopal Inlier, Bhander Group, Vindhyan basin.

INTRODUCTION

The Vindhyan basin (Fig. 1) is a large Proterozoic sedimentary basin located in central peninsular India that outcrops over an area of 104,000 km^2, with additional area covered by the Deccan Traps and Indo-Gangetic alluvium (Venkatachala et al. 1996). Geographically, the basin lies between the gneiss and granite of the Archean (>2.5 Ga) Aravalli province to the north and Bundelkhand to the east (Mazumdar et al. 2000) and Deccan flood basalts to the south. Across the modern day Aravalli mountains is the 54,000 km^2 and 750–771 Ma Malani Igneous Province, unconformably overlying sediments of the Neoproterozoic–Cambrian Marwar Supergroup (Raghav et al. 2005). Sedimentary units in the Vindhyan basin are primarily represented by shallow marine facies along with distal shelf to deep-water sediments in the Lower Vindhyan and Lower Rewa, and can be subdivided into four Groups: The Semri (or "Lower Vindhyan" sequence), Kaimur, Rewa and Bhander forming the "Upper Vindhyan" sequence (Chaudhuri et al. 1999). The Lower Vindhyan and Upper Vindhyan units are separated by multiple unconformities of undetermined duration (Bose et al. 2001). The lower Vindhyan units are collectively designated as the Semri Group. The Semri is noteworthy for good age control from Pb–Pb ages of carbonate units, as well as precise U–Pb ages derived from zircon separated from volcanoclastic strata (Ray et al. 2002; Rasmussen et al. 2002a).

The paucity of high quality paleomagnetic data hinders the reconstructions and the refinement of Apparent Polar Wander paths (APW) of Rayner and Mawson cratonic blocks (East Antarctica), Australia, Madagascar, the Seychelles, Sri Lanka and India for Mesoproterozoic supercontinent of Rodinia as well as their subsequent coalescence in the continent of Gondwana following the Neoproterozoic breakup of Rodinia time interval (Meert and Powell, 2001). Research on the Indian subcontinent provides an important window into this problem as it is both accessible and contains targets of the appropriate age. The Vindhyan basin located in the central peninsular region of India, provides a promising area to conduct the necessary paleomagnetic studies due to the long depositional history recorded in the basin, limited deformation and unmetamorphosed nature of the rock throughout the basin. The data generated from this study will aid in constraining the position of India during the Meso-to Neoproterozoic interval, generate points that can be used in a Proterozoic APW path for India, and test the hypotheses on breakup of Rodinia and assembly of East Gondwana.

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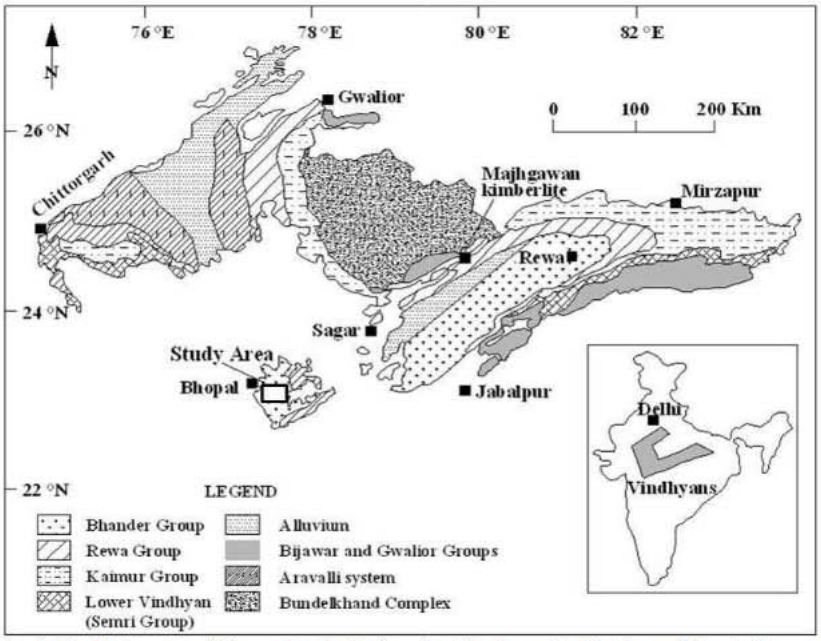


Fig.1. Geologic map of the region including the Vindhyanchal basin and the study area.

GEOLOGY AND SAMPLING

The outcrop area of the Vindhyan basin is divided into two terrains: the Rajasthan terrain in the present day west region and the Uttar Pradesh–Madhya Pradesh–Bihar region in the eastern sector of the modern day regional extent (Mitra, 1996). Stratigraphically, Vindhyan basin can be divided into two sequences: The Lower Vindhyan Sequence formed by Semri Group and the Upper Vindhyan Sequence sub-divided into the Kaimur, Rewa and Bhander Groups, respectively (Chaudhari et al. 1999). Generally the Vindhyan beds dip less than 10 degrees, they are practically undisturbed and sub-horizontal in the sampling area.

The Bhopal and Indore inliers also expose Vindhyan Supergroup sedimentary rocks much away from the main basin in Son Valley and southeast of Rajasthan. Palaeomagnetic investigations were taken up on Bhopal Inlier sediments with a view to understand the tectonic history and correlation of their contemporaneity.

A total of 50 oriented block samples from 10 sites representing sandstones were collected from Bhopal Inlier. Sample orientation was performed in the field using Brunton magnetic and SUN compasses, and solar readings were used to correct any magnetic deflections and local declination deviations.

PREVIOUS WORK

Palaeomagnetism

The Vindhyan Basin has been the subject of several paleomagnetic studies. Palaeomagnetic data (D = 357°, I = 31°) were acquired from the Upper Vindhyan Kaimur series in Central India (Sahasrabudhe and Mishra, 1966, as quoted by McElhinny, 1968). These authors originally assigned a Cambrian age (540-580 Ma.) to their samples but the radiometric dating (Vinogradov et al. 1964; Tugarinov et al. 1965; Crawford and Compston, 1970) proved the age of the Kaimur series to be much older between 900 and 1200 Ma. This Upper Vindhyan Kaimur result seems to be no longer representative for the Cambrian of India. Paleomagnetic results from sandstones of the Bhander series, reported by Athavale et al. (1972) were different from the present results because he treated Upper Bhander and Rewa sandstone samples to alternating fields up to 80 mT and thermal demagnetization steps up to 600°C. All these experiments were carried out on a classical instruments like Astatic magnetometer and Helmholtz coil system AF and thermal demagnetizers. It is believed that Astatic magnetometers are not suitable for measuring NRM of weakly magnetized sediments and moreover, after demagnetization it is further difficult to obtain the magnetic signatures as the intensity drops drastically. These investigations yielded mean direction of Dm = 48°, Im = –19°, k = 200, α_{95} = 5.7° for Bhander Group and mean direction of D = 32°, I = –37°, k = 15, α_{95} = 13.7° for Rewa Group. These results yielded a paleomagnetic pole of 32° N and 199° E for the Bhander and 35 °N and 222 °E for the Rewa respectively. Klootwijk (1973) analyzed 43 cores from seven sites in Rajasthan by applying progressive alternating field (AF) and thermal treatments on the samples, and generated a combined site mean of D = 207.5°, I = +9.5°, k = 137.5, α_{95} = 5.5° and a paleo-magnetic pole at 51 °N and 214 °E. A later study conducted by McElhinny et al. (1978) expanded sampling into the lower sandstone of the Bhander and included one site in the Rewa Group. In all, seven sites were sampled and subjected to a thermal demagnetization treatment. Three vectors were identified: a viscous component aligned with the present day field, a Tertiary overprint associated with the Deccan Traps emplacement, and a primary direction evident above 600–665 °C. This primary direction, averaged for the seven Bhander–Rewa sites gives D= 203.4°, I = +8.1° (k = 36.5, α_{95} =11.2°) with a Rewa VGP at 45.0 °N, 191.3 °E and Bhander paleomagnetic pole at 51.3 °N, 222.7 °E (McElhinny et al. 1978).

2 of 7 12/03/2013 12:30 PM

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Age control on the Upper Vindhyan sequences is more complex. Ages from within the Upper Vindhyan sedimentary units lack consistency and reliability. The Kaimur sandstone has a reported K–Ar age on authigenic glauconite of 910±39 Ma. (Vinogradov et al. 1964) that is too young in light of the age of the Majhgawan kimberlite that intrudes the Kaimur series. Fission track ages from the Govindgarh sandstone (upper Rewa Group) yielded an age of 710±120Ma (Srivastava and Rajagopalan, 1988). Recent Pb–Pb dating of Bhander Group carbonates produced an unreliable age of 650±70 Ma, however, this age appears consistent with samples taken from the Bhander–Lakheri limestone that yields a $^{87}Sr/^{86}Sr$ value consistent with global values near 650 Ma (Ray et al. 2003). Further isotopic studies of the Bhander limestone $^{87}Sr/^{86}Sr$ values indicate a 750 Ma age when compared to global curves for the Neoproterozoic (Ray et al. 2003). To these uppermost Upper Vindhyan beds a probable early Palaeozoic age has been assigned (Ahmad, 1958; Gansser, 1964; Krishnan, 1968; Crawford and Compston, 1970).

Non-isotopic methods of dating the Upper Vindhyan units have also been attempted, with equivocal results. Possible Ediacara fauna fossils of nine coelenterate genera (*Tribachidium*, *Eoporita*, *Kaisalia*, *Cyclomedusa*, *Ediacaria*, *Nimbia*, *Paliella*, *Medusinites*, *Hiemaloria*), one proto-arthropod (*Spriggina*) and several unidentified taxa have been described in the Lakheri and Sirbu formations of the Bhander Group and would indicate an Ediacaran age (<635 Ma) for the Bhander Group (De, 2003, 2006). This fauna is useful both for the biostratigraphic age constraint as well as for correlations with other Ediacara sites worldwide (De, 2006).

Palaeomagnetic directions obtained from the Bhander and Rewa appear to correlate with late Neoproterozoic to Cambrian data from Pakistan (McElhinny et al. 1978). These correlations are suspect due to significant rotations in the Salt Range (Klootwijk et al. 1986). Similarities between the Bhander–Rewa paleomagnetic pole and those of other Gondwana cratons have been drawn as well. Many publications (e.g. Meert, 2001; Powell and Pisarevsky, 2002) place the Bhander and Rewa poles on the late Neoproterozoic to Cambrian Apparent Polar Wander path for Gondwana, assuming a ±550 Ma age for the Upper Vindhyan and comparing the poles to the 547 Ma Sinyai dolerite pole (Meert and Van der Voo, 1996) or the 645–635 Ma (Condon et al. 2005; Kendall et al. 2006). In a recent study Malone et al. (2008) have emphasized the age of the upper Vindhyan sequences as 1000 Ma, which, is the closer age for the Purana basins? They have concluded this based on the detrital zircon analysis of upper Bhander sandstones. Whereas no zircon dating is done on Bhopal Inlier sandstone. Palaeomagnetic pole position obtained from this study is more similar to the Malani igneous suite results by Gregory et al. (2009).

METHODS AND RESULTS

The samples were cut into cylindrical specimens (size 25 mm diameter x 22 mm height) of relatively uniform volume in the laboratory. Sample susceptibility was measured using Bartington Susceptibilty Meter (Model MS-2) Instrument. Natural Remanent Magnetization (NRM) measurements were carried out on Molspin minispin Magnetometer (Magnetic Measurements, U.K.) and sensitivity of the Magnetometer is 2.5 x 10^{-8} EMU.cm^{-3} (2.5 x 10^{-5} A/m). The obtained directions from these samples reveal some scatter with upward and downward inclinations. Pilot samples were selected for preliminary demagnetization and a sequence of demagnetization steps was chosen based on these preliminary results. Thermal demagnetization was carried out using thermal demagnetizer (model MMTD-80, Magnetic Measurements, U.K) and for AF demagnetization treatment, a Molspin AF demagnetizer (Magnetic Measurements, U.K) is used. The resulting data was analyzed using principle component analysis of a best fit line using REMA 4 software.

The samples collected for this study are sandstones containing orthoclase, feldspars, and few specks of magnetite and heamatite minerals with quartz being the dominant mineral.

The average intensity for the Bhopal Inlier sediments is 7.34 mA/m. AF and thermal demagnetization studies on these samples reveal good grouping with steep downward inclination as shown in Fig.2. Figure 3 show the Stereo plot of the Characteristic Remanent Magnetic (ChRM) directions for Bhopal Inlier sediments. This ChRM directions reveal a mean direction of Dm = 357°, Im = 58°, k = 17.69 and α_{95} = 16.38°. Virtual Geomagnetic Pole (VGP) of each individual site has been calculated. The average VGP of this formation gives a Pole position of 73.88 °N and 69.04 °E (dp=17.2, dm=21.72) with a colatitude (λ_m) = 38.7° N.

DISCUSSION

Age control on Vindhyan sedimentation is still the subject of considerable controversy as of the ages of the other Purana basins (Patranabis-Deb et al. 2007; Gregory et al. 2006). The Semri Group is separated from the Upper Vindhyan by a basin wide unconformity between the Rohtas limestone and the overlying Kaimur Group. A thin

3 of 7 12/03/2013 12:30 PM

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Table 1. Paleomagnetic results from Bhopal Inlier and other studies on Malani Igneous Province

Site name	*n*/N	Mean Dec.	Mean Inc.	k	α_{95}	VGP latitude	VGP longitude
This study							
Site1	5/5	358	47	20.46	17.32	84.8°N	57.8°E
Site2	4/5	354	28	67.86	11.23	79.9°N	292.6°E
Site3	5/5	7	52	23.90	15.97	78.8°N	109.6°E
Site4	5/5	359	71	51.06	10.81	57.8°N	76.5°E
Site5	5/5	339	81	786.41	2.73	39.5°N	69.5°E
Site9	5/5	354	66	44.79	11.56	64.5°N	68.3°E
Combined mean	**6 sites**	**357**	**58**	**17.69**	**16.38**	**73.9°N**	**69.0°E**
Gregory et al. (2009)							
1434 (normal)	23/27	351	73	1389.55	2.7	56.9°N	64.3°E
1434 (reverse)	3/3	195	-60	234.79	8.1	70.2°N	108.8°E
1435	6/8	350	62	244.78	4.3	70.5°N	49.8°E
1436	9/9	355	58	256.88	3.8	75.9°N	57.7°E
Combined mean	**4 Dikes**	**359**	**64**	**91.2**	**9.7**	**70.2°N**	**70.1°E**
Torsvik et al. (2001a, b)							
1	5	39	71	188.9	5.6	49.7°N	106.8°E
3	13	17	52	678.2	1.6	73.8°N	136.7°E
4	13	313	72	51.7	5.8	44.5°N	39.0°E
5	6	356	64	511.1	3.0	70.1°N	64.7°E
6	11	25	47	178.9	3.4	68.0°N	152.8°E
8	16	354	59	322.0	2.1	74.6°N	54.9°E
10	4	340	64	51.6	12.9	63.9°N	39.5°E
13	5	58	74	109.0	7.3	38.0°N	104.7°E
14	7	13	62	201.0	4.3	69.5°N	99.6°E
Combined mean	**9**	**09**	**65**	**26.35**	**9.6**	**67.8°N**	**72.5°E**
Klootwijk (1975)							
RI-2	6	4	63	133.0	6.5	72.2°N	82.2°E
RI-3	4	24	83	297.5	5.3	39.6°N	80.6°E
RI-7	4	340	45	55.0	12.5	72.1°N	349.0°E
RI-10	5	347	52	162.0	6.0	76.4°N	15.4°E
RI-12	3	337	55	316.0	7.0	68.2°N	12.7°E
RI-13	4	345	81	490.5	4.0	42.5°N	67.1°E
Combined mean	**6**	**347**	**64**	**24.0**	**14.1**	**68.2°N**	**47.9°E**

n = samples used; *N*= samples collected; k= kappa precision parameter; α_{95} = circle of 95% confidence about the mean; VGP = Virtual Geomagnetic Pole

shale unit marks the transition into the Bhander Group. The Bhander Group contains the only major carbonate unit in the upper Vindhyan system, a unit containing stromatolites, ooids, and micritic layers known as the Bhander or Lakheri limestone (Bose et al. 2001). The overlying lower Bhander sandstone marks a transition from shallower marine, to sometimes fluvial, typical of the Bhander Group (Bose et al. 2001). The Sirbu shale overlies the lower Bhander sandstone, and is in turn overlain by the upper Bhander sandstone. The best age estimates come from the Majhgawan kimberlite, that intrudes the Lower Vindhyan and into the Baghain sandstone (Kaimur Group – Upper Vindhyan) near Panna. A study conducted by Malone et al. (2008) noted that the youngest population of zircons from the Upper Bhander is older than 1000 Ma. This observation, coupled with the similarity in paleomagnetic directions from the Upper Vindhyan and Majhgawan kimberlite led to conclude that Upper Vindhyan sedimentation was completed by ~1000 Ma. This result is consistent with recent data from another study on the Purana basins to the south (Patranabis-Deb et al. 2007). Crawford and Compston (1970) determined an age of 745±10 Ma (Rb-Sr method) for the Malani granites and rhyolites. Dhar et al. (1996) determined an age of 723±6 Ma for the Malani granites and rhyolites. The Malani Igneous Suite has been studied paleomagnetically by Athavale et al. (1963), Klootwijk (1975) and Torsvik et al. (2001). These studies yielded similar paleomagnetic results, but statistically significant fold tests were not obtained except Torsvik et al. (2001a,b) and thus could not constrain the relative magnetisation age (pre- or post-fold).

Nantua deposits (748 Ma) of glaciogenic origin, corresponding to the Pokhran boulder bed and subsequent

4 of 7 12/03/2013 12:30 PM

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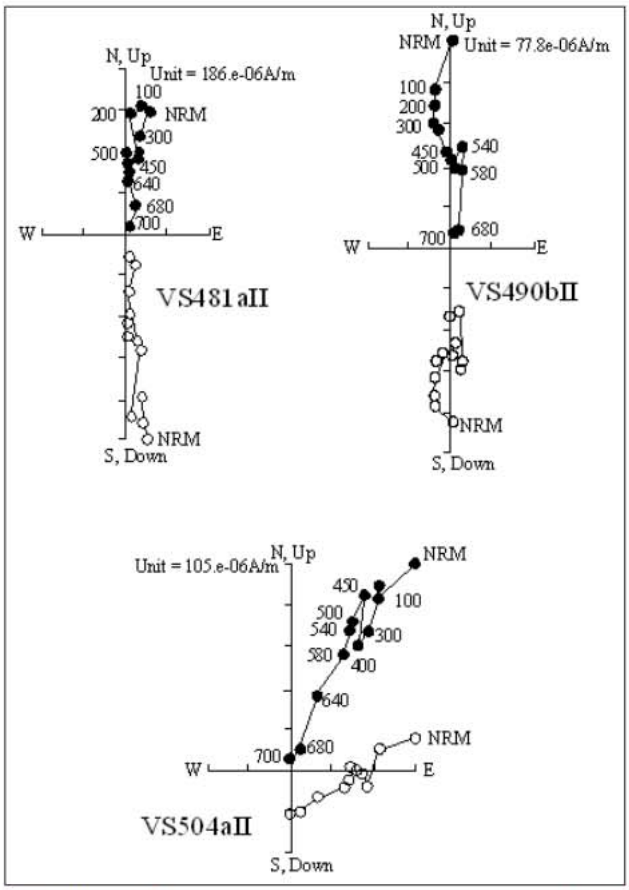


Fig.2. Vector End Point diagram during progressive thermal demagnetization for Bhopal Inlier sample showing univectorial behavior. All the steps are in °C.

desiccation exemplified by carbonate mainly dolomite and phosphate deposits correlatable with Hanseran evaporites of Marwar basin. All these similarities have led Kochhar (2007, 2008) to suggest that the Yangtze craton (YC) of south China was attached to the Trans-Aravalli Block (TAB) of NW Indian shield in the assembly of the Malani Supercontinent. It has been suggested by Kochhar (2001 a, b) that the position of Indian subcontinent in the assembly of a Late Proterozoic Supercontinent should be viewed in

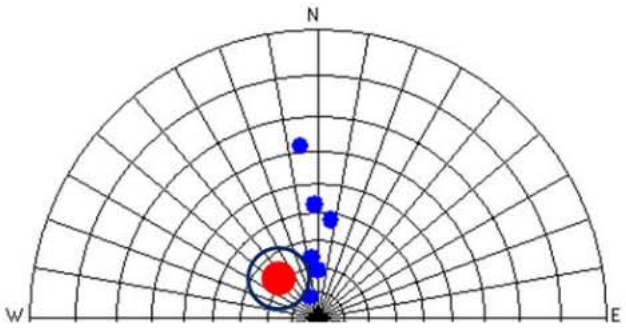


Fig.3. Stereo plot of Site mean directions.

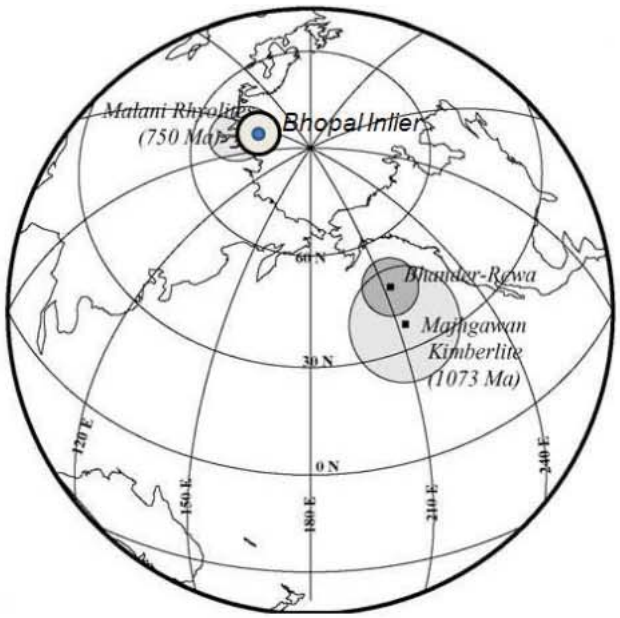


Fig.4. Virtual geomagnetic pole for Bhander Rewa Groups, Malani Igneous province, Majhgawan kimberlites and Bhopal inlier.

terms of three major blocks with different magmatic, metamorphic and tectonic histories, and not as a single entity.

Kochhar (1996) proposed that 730 Ma marked a major tectonomagmatic event of widespread intraplate, anorogenic magmatism represented by alkali granites and cogenetic acid volcanics in the northwestern Indian shield, the Trans-Aravalli block (TAB) [Malani Igneous Suite], central Iran, Nubian-Arabian shield, Somalia and Seychelles. In view of the commonality of crustal stress pattern, rifting and thermal regime, which, gave rise to this anorgenic magmatism. Based on this magmatism, it has been proposed that all these micro continents formed a supercontinent termed here the Malani Supercontinent. Paleomagnetic data also support the existence of such a Supercontinent around c. 750 Ma. Paleomagnetic data from Seychelles and Malani Igneous Suite (MIS) yield local paleolatitude of 30° N and 40° N, and a new Malani Igneous Suite - Seychelles fit, places Seychelles only 600 km apart from MIS during the Mid-Proterozoic.

The Majhgawan intrusion into the Kaimur Group provides an important reference as a cross cutting intrusion at 1073.5±13.7 Ma. The VGP generated from the Majhgawan kimberlite, the Harohalli dykes paleomagnetic pole, and the paleomagnetic pole from the Malani Igneous province provide the best temporally constrained paleomagnetic information covering the suspected period of deposition of the Bhander and Rewa Groups.

5 of 7 12/03/2013 12:30 PM

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CONCLUSION

Palaeomagnetic data from Bhander Inlier sediments are of excellent quality and resembles previous palaeomagnetic results on Malani igneous Suite (MIS). This study provides a refined paleomagnetic pole for the Upper Vindhyan sequence of the Vindhyan basin at 74° N, 69° E (α_{95} = 16.38°). This paleomagnetic pole correlates well with the Virtual Geomagnetic Pole generated from the Malani Igneous Suite (MIS) by Gregory et al. (2009) combined with other studies on MIS lies at 67.8 °N, 72.5 °E (A_{95} = 8.8°). By assigning this age to the Bhopal Inlier sediments several interesting possibilities can be considered. The Bhander and Rewa Groups are not well constrained by reliable direct age dates. The high-stability magnetisation (771–751 Ma) of MIS yields a local paleolatitude of 41 °N (+8/"7) when combined with normal polarity sites of Klootwijk (1975). Bhopal Inlier sediments also recorded the similar Paleolatitude of 39 °N. MIS and the Seychelles micro-continent formed the western margin of the Rodinia Supercontinent at 750–755 Ma. Paleomagnetic data imply that the Seychelles formed at ca. 30° N, and we presume that the Neoproterozoic igneous rocks of Malani (NW India), Bhopal Inlier, the Seychelles and northern Madagascar probably constituted an Andean type arc, formed on the western margin of East Gondwanaland, above an east-dipping subduction zone.

The simplest interpretation for the age of the Bhander and Rewa Group is that the pole is close to the age of Majhgawan kimberlite (i.e. between 1000 and 1070 Ma). Whereas, our present Paleomagnetic pole fits very well with that of MIS pole. Therefore, we conclude that the Bhanders of Bhopal Inlier may be older than what earlier thought. These data support the idea that East Gondwana did not coalesce until the end of Neoproterozoic–Cambrian transition, as suggested by Meert et al. (1995), Powell and Pisarevsky (2002) and Meert (2003).

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6 of 7 12/03/2013 12:30 PM

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7 of 7 12/03/2013 12:30 PM